InTRACt · Interplay of Text, Reader And Context

The language on the screen: children's reading in a multilingual story app

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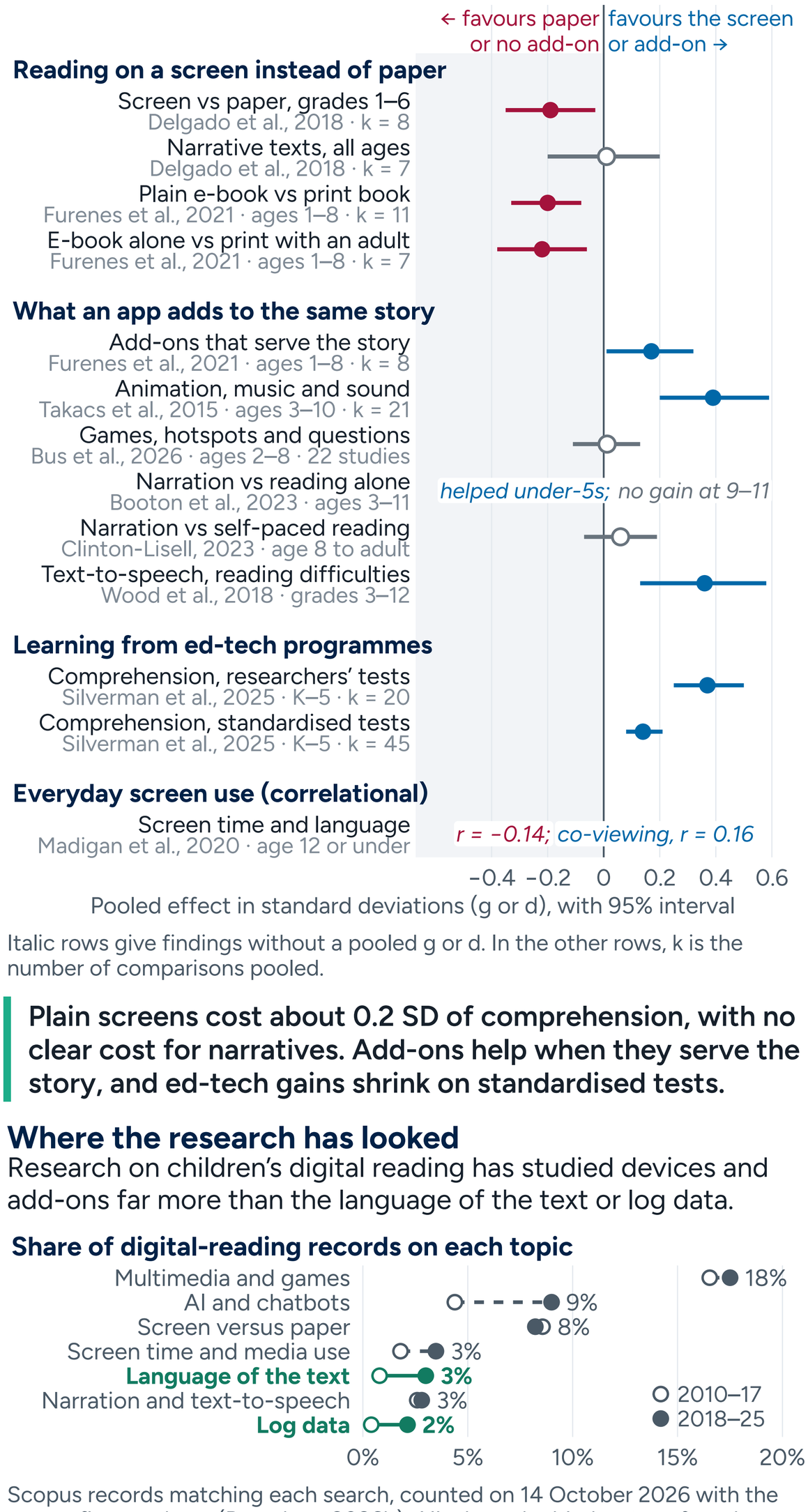
DEPARTMENT OF EDUCATION · UNIVERSITY OF OXFORD

Research has asked whether screens help or hurt children's reading. We ask how the language of on-screen stories shapes reading too.

WHAT WE KNOW

Screens, apps and children's reading

In 2025, more than one in five UK 8- to 18-year-olds read fiction on a screen at least monthly (Picton & Clark, 2026).



Italic rows give findings without a pooled g or d. In the other rows, k is the number of comparisons pooled.

Plain screens cost about 0.2 SD of comprehension, with no clear cost for narratives. Add-ons help when they serve the story, and ed-tech gains shrink on standardised tests.

Where the research has looked

Research on children's digital reading has studied devices and add-ons far more than the language of the text or log data.

Scopus records matching each search, counted on 14 October 2026 with the scopusflow package (Bernabeu, 2026b). AI's share doubled, yet we found no meta-analysis of generative AI and children's reading.

WHAT WE WILL TEST

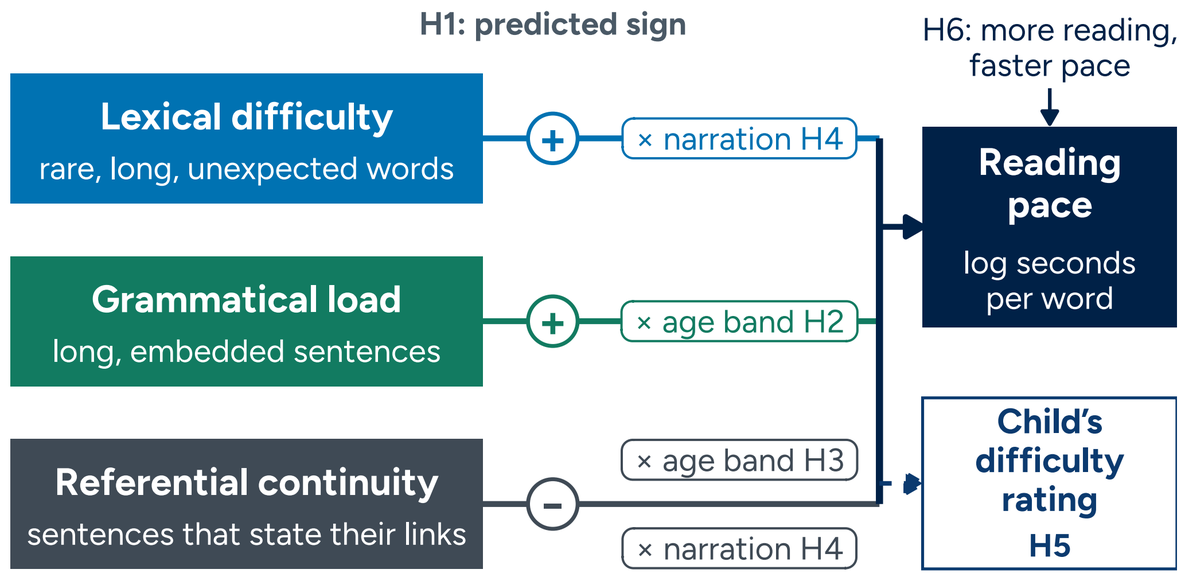
0 of 1,536 records on children's digital reading matched both our language-of-the-text and log-data searches. This study brings the two together.

Let's Story: AI stories, logged at scale

In this feature of a children's app, a generative model writes stories from the character, setting, plot and theme a child picks (Ratner et al., 2025). In its first month, *Let's Story* logged 419,700 story sessions in 18 languages (Ratner & Murphy, 2026).

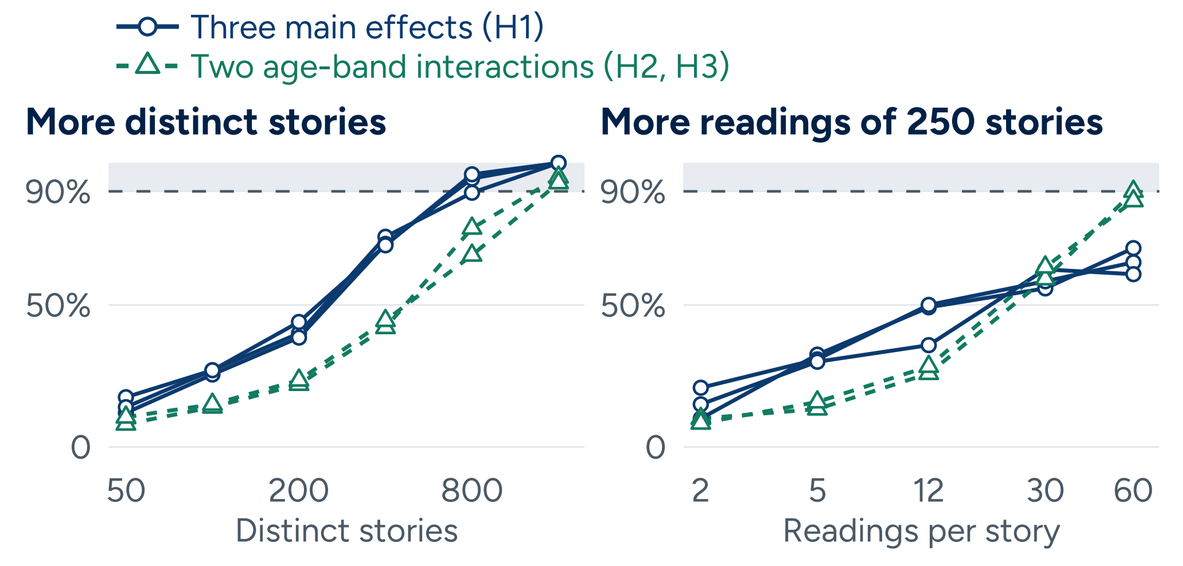
Six confirmatory hypotheses

We test whether text features that slow or speed reading in the laboratory do the same in the app.



A plus sign predicts slower reading (more seconds per word), a minus sign faster reading. H2 and H3 are two-sided because rival accounts predict opposite signs. The analysis covers the app's 7–8 and 9+ age bands, in ten of its languages.

Precision comes from stories, not sessions



Chance that a 0.10 SD effect is judged non-negligible (95% interval outside ±0.05 SD), simulated with the pilotr package (Bernabeu, 2026a). A main effect needs 630–740 distinct stories, or about 930 under the registered Bayes-factor rule. Assumed: one reading per child, 9% of the variance between stories and 20% between children, equal age bands, independent text measures, and no language terms, covariates or priors.

Effects that hold in the app could inform how AI stories are pitched to each age band, though faster reading is not better understanding. Effects that fade may reflect home reading conditions or the measure as much as the theory.

A Registered Report on sealed data

1. Peer review before any outcome is seen
2. Logs released only after acceptance
3. Text measures fixed blind to outcomes
4. Decision rules set in advance

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References, protocol and code

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